`Semigroups` — Zero to Hero — Worksheet

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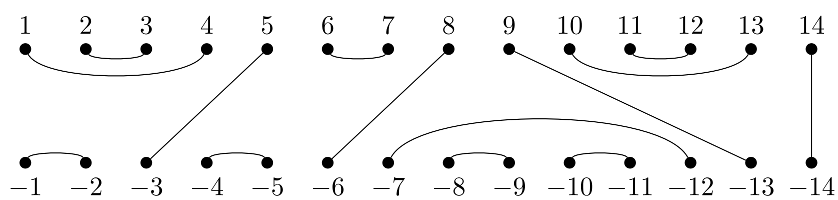
If you've never used GAP before you might want to look at:
https://carpentries-incubator.github.io/gap-lesson/
before attempting any of the problems on this sheet.

This worksheet contains some exercises to be completed in `GAP` with the use of the `Semigroups` package as part of the GAP workshop session of NBSAN 40. The problems have been written in a deliberately terse way to encourage attendees to experiment with the package and explore its associated manual: https://semigroups.github.io/Semigroups/doc/chap0_mj.html.

As well as inspecting the package manual, please also feel free to ask questions to James, Reinis or Joe!

1. Let J_n denote the Jones monoid of degree n, and

$x =$ 

be a bipartition of degree 14.

(a) Find a collection of bipartitions $\{x_1, x_2, \ldots, x_j\}$ such that the tensor product $x_1 \otimes x_2 \otimes \cdots \otimes x_j$ is equal to x. What is the largest possible value of j? For the definition of the tensor product, see the documentation of the function `TensorBipartition`.

(b) How many elements does J_{14} contain?

(c) How many idempotent elements does J_{14} contain?

(d) How many idempotent elements does the Green's $\mathcal{D}$-class of x in J_{14} contain?

(e) How many idempotent elements e, for which 7 is in the domain of e, does the Green's $\mathcal{D}$-class of x in J_{14} contain?

(f) How many irreducible idempotent elements $\hat{e}$, for which 7 is in the domain of $\hat{e}$, does the Green's $\mathcal{D}$-class of x in J_{14} contain?

2. Recall that the commuting graph of a semigroup S is the graph with nodes the elements of S and an edge (x, y) if $xy = yx$ holds. Let T_n be the full transformation monoid on $n \leq 5$ points. Show that the clique numbers of the commuting graph of T_n are 2^{n-1}.

3. Let S be the semigroup defined by the presentation

$$\langle a, b, c, d, e, f, g \mid abcd = aaaeaa, ef = dg \rangle.$$

   (a) Show that S is infinite.

   (b) Create a homomorphism from the free semigroup $\{a, b, c, d, e, f, g\}^+$ to S.

   (c) Partition the first 1000 elements of $\{a, b, c, d, e, f, g\}^+$ so that words belong to the same part if and only if they represent the same element of S.

4. Is the semigroup defined by the presentation

$$\langle x_2, \ldots, x_n \mid x_i^2 = (x_i x_j)^3 = (x_i x_j x_k)^4 = 1 \quad i, j, k \text{ distinct} \rangle.$$

   the symmetric group?

5. Determine which of the relations in the presentation are redundant and which are not:

$$\langle a, b \mid a^4 = 1, b^2 = b, ba^3ba = a^2(ab)^2, (ba^2)^2 = (a^2b)^2, (ba)^2a^2 = aba^3b, a(ab)^4 = (ab)^4 \rangle$$

   What is the minimal set of the relations in this presentation that define the same monoid?

6. Let S be the Catalan monoid of degree 3.

   (a) Draw the egg-box diagram of S.

   (b) Draw the left and right Cayley graphs of S.

   (c) Show that S has two non-trivial non-universal non-Rees congruences.

   (d) Show that the lattice of left and right congruences of S are isomorphic.

   (e) What are the maximal subsemigroups of S? Show that none of the maximal subsemigroup is isomorphic to any of the others.

7. Let S be the dual of the full transformation monoid on 5 points. Find a transformation representation of S on 32 points.

   (a) Draw the egg-box diagram of S.

   (b) Draw the left and right Cayley graphs of S.

   (c) Show that S has two non-trivial non-universal non-Rees congruences.

   (d) Show that the lattice of left and right congruences of S are isomorphic.